

Quiz (Transactions and Locks Part I)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You will be allowed to use a computer and access the internet, but you will not be allowed to connect to postgres and run postgres commands.

Problem 1:

Session 1	Session 2
1 CREATE TABLE t (a INT);	1
2 CREATE TABLE u (a INT UNIQUE);	2
3 BEGIN;	3
4 LOCK TABLE t IN EXCLUSIVE MODE;	4
5	5 BEGIN;
6	6 LOCK TABLE u IN
7	7 SHARE MODE;
8 LOCK TABLE u IN ROW EXCLUSIVE MODE;	8
9	9 LOCK TABLE t IN
10	10 ROW EXCLUSIVE MODE;
11 ABORT;	11
	12 COMMIT;

Problem 2:

Session 1	Session 2
1 CREATE TABLE t (	1
2 a INT NOT NULL	2
3);	3
4 BEGIN ISOLATION LEVEL	4
5 READ COMMITTED;	5
6	6 BEGIN;
7	7 INSERT INTO t VALUES (6);
8 INSERT INTO t VALUES (5);	8
9 INSERT INTO t VALUES (6);	9
10 SELECT count(*) FROM t;	10
11	11 INSERT INTO t VALUES (5);
12	12 COMMIT;
13 ROLLBACK;	

Problem 3:

Session 1	Session 2
1 CREATE TABLE t (a INT);	1
2	2 BEGIN ISOLATION LEVEL
3	3 READ UNCOMMITTED;
4	4 SELECT count(*) FROM t;
5 BEGIN;	5
6 INSERT INTO t VALUES (1);	6
7 INSERT INTO t VALUES (2);	7
8	8 SELECT count(*) FROM t;
9 INSERT INTO t VALUES (3);	9
10 COMMIT;	10
	11 SELECT count(*) FROM t;
	12 COMMIT;
	13 SELECT count(*) FROM t;

Problem 4:

Session 1	Session 2
1 CREATE TABLE t (a INT);	1
2 INSERT INTO t VALUES (1);	2
3 INSERT INTO t VALUES (2);	3
4 INSERT INTO t VALUES (3);	4
5	5 BEGIN;
6	6 SELECT count(*);
7 BEGIN;	7
8 UPDATE t SET a=6 WHERE a=1;	8
9	9 DELETE FROM t WHERE a=1;
10	10 COMMIT;
11 INSERT INTO t VALUES (1);	11
12 ROLLBACK;	12
	13 SELECT count(*) FROM t WHERE a=1;